

A Mixed-Base Counterexample to Stanley's Rationality Question:

Exact Coefficient Counts, a Two-Base Rationality Classification, and Non-D-Finiteness via a Natural Boundary

Exact coefficient counts, irrational thresholds,
finite-quotient rigidity, and a natural boundary

Research report prepared for Vladimir Reshetnikov

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Abstract

Richard Stanley asked whether a positive integer sequence tending to infinity and having a rational generating function necessarily produces rational generating functions for the numbers of coefficients equal to a fixed integer in its associated finite products. We give a counterexample using the interleaved sequences 2^j and 3^j . For $P_N(x) = \prod_{i=1}^N (1 + x^{G_i})$, with $G_{2j+1} = 2^j$ and $G_{2j+2} = 3^j$, the number a_n of coefficients equal to 1 in P_{2n} is

$$a_n = 2^{n - \lfloor (n+1) \log_3 2 \rfloor} + \mathbf{1}_{\{n=1\}}.$$

The proof turns polynomial coefficients into interval-covering multiplicities and then counts gaps created by ternary carries. Two independent algebraic routes defeat rationality: an elementary finite-field argument that reduces an assumed recurrence modulo a prime, and a finite-quotient rigidity theorem that needs no prime and yields the strictly stronger conclusion that the counts are not P-recursive. Neither route makes any transcendence assumption about the exponential growth constant. A separate analytic argument proves a natural boundary — once by a direct Cesàro computation on the actual coefficient profile, and once by a general theorem covering $\sum_n q^{\lfloor n\theta \rfloor} t^n$ for every irrational θ and every $q > 1$ — hence non-D-finiteness and the absence of polynomial-coefficient linear recurrences. Finally, for a two-base family with digits $0, \dots, a-1$, we prove a five-way equivalence: rationality, D-finiteness of either the even section or the full series, and multiplicative dependence of the bases are all the same condition. Exact-integer verification code, sequence data, and reproducible figures accompany the proofs.

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1 The problem and the result

In a MathOverflow question dated September 23, 2022, Richard Stanley considers a sequence of positive integers $(G_N)_{N \geq 1}$ satisfying

$$G_N \longrightarrow \infty, \quad \sum_{N \geq 1} G_N t^N \in \mathbb{Q}(t). \quad (1.1)$$

For a fixed integer $r \geq 1$, put

$$P_N(x) = \prod_{i=1}^N (1 + x^{G_i} + \cdots + x^{rG_i}), \quad P_0(x) = 1. \quad (1.2)$$

For fixed $k \geq 1$, let $v_k(N)$ be the number of exponents $s \geq 0$ for which $[x^s]P_N(x) = k$. The question asks whether $\sum_{N \geq 0} v_k(N)t^N$ is rational and, if not universally, which sequences G_N have this property [1].

The present report gives a negative answer to the universal assertion, already for $r = k = 1$. The source question requires divergence to infinity, *not monotonicity*. That distinction matters: the counterexample below is not monotone.

1.1 An explicit counterexample

For every $j \geq 0$, define

$$G_{2j+1} = 2^j, \quad G_{2j+2} = 3^j. \quad (1.3)$$

Thus the beginning of the exponent sequence is

$$1, 1, 2, 3, 4, 9, 8, 27, 16, 81, 32, 243, \dots$$

Its generating function is visibly rational:

$$\sum_{N \geq 1} G_N t^N = \frac{t}{1-2t^2} + \frac{t^2}{1-3t^2} = \frac{t + t^2 - 3t^3 - 2t^4}{1-5t^2+6t^4}. \quad (1.4)$$

Equivalently,

$$G_{N+4} = 5G_{N+2} - 6G_N \quad (N \geq 1). \quad (1.5)$$

Both parity subsequences tend to infinity, so (1.1) holds. Write

$$v(N) = v_1(N), \quad a_n = v(2n), \quad b_n = v(2n+1), \quad A(z) = \sum_{n \geq 0} a_n z^n, \quad F(t) = \sum_{N \geq 0} v(N)t^N. \quad (1.6)$$

The variable x always records polynomial exponents; z and t record product lengths. In particular, A is a section of F , not a specialization of the product variable x .

Theorem 1.1 (Counterexample and analytic strengthening). *Set $\alpha = \log_3 2$ and $R = 2^{\alpha-1}$. For the exponents (1.3),*

$$a_n = 2^{n - \lfloor (n+1)\alpha \rfloor} + \mathbf{1}_{\{n=1\}} \quad (n \geq 0). \quad (1.7)$$

Neither $A(z)$ nor $F(t)$ is rational. In fact, $|z| = R$ is a natural boundary for A , and $|t| = \sqrt{R}$ is a natural boundary for F . Consequently both functions are non- D -finite and transcendental over their rational function fields, and neither coefficient sequence is P -recursive.

A second, independent algebraic route to the same conclusion — and to more — is given in Section 5. It replaces reduction modulo a prime by a rigidity theorem for sequences whose adjacent quotients lie in a fixed finite set, and it rules out polynomial-coefficient recurrences, not merely constant-coefficient ones.

Theorem 1.2 (Two-base classification, informal form). *Fix integers $2 \leq a < q$ and interleave the exponents a^j and q^j with the digit polynomial $1 + y + \cdots + y^{a-1}$. Then the even-section series, the full series, rationality, and D-finiteness all stand or fall together: each of the four analytic properties holds if and only if a and q are multiplicatively dependent. The precise statement is Theorem 9.1.*

For $a = 2$ the dependence condition reads “ q is a power of two,” so every other $q \geq 3$ gives a counterexample already with the binomial factors $1 + x^{G_i}$.

The elementary counterexample is complete by Section 4. Section 5 gives the rigidity route and the non-P-recursiveness. Sections 6–8 establish the exact floor formula, the fluctuation law, and the analytic conclusions. Section 9 proves the classification.

1.2 Provenance of this report

This report is a merge of two independently produced research reports on the same MathOverflow question, originally distributed as the directories `mixed-base-rationality-classification` and `mixed-base-natural-boundary`. The shared enumerative core — the interval-coverage identity, the carry-gap distribution, the exact rectangular count, and the floor formula — was obtained by the same argument in both, and is proved exactly once below, in the notation of the first report.

Where the two reports genuinely diverge, both arguments are preserved and marked as such. There are three such places. Section 4 gives the first report’s elementary reduction modulo a prime, and Section 5 gives the second report’s finite-quotient rigidity theorem; Section 5 opens with a paragraph saying what each buys. Section 4 also retains the first report’s companion-matrix argument for even sections, while Lemma 5.4 supplies the second report’s general decimation lemma. Section 8 proves the natural boundary twice, by a direct computation on the actual profile (Theorem 8.2) and by a general irrational-floor theorem (Theorem 8.3); the two resulting radial constants are reconciled explicitly in Remark 8.5. The companion note `short_proof.md` mirrors the same fork in compact form.

1.3 Source status and scope of the claim

The source page displayed no posted answer when inspected on September 19, 2026. Targeted searches for the question title, its identifier, and mixed-base counterexamples did not reveal an existing resolution. This is a limited literature audit, not a proof of priority. The mathematical assertions here are supported by the proofs below; the report has not been independently refereed or formally verified in a proof assistant.

Stanley’s related work on rational generating functions for moments of polynomial coefficients supplies useful context [4], as does the study of linear recurrences arising from Stern’s diatomic array [3]. Those moment questions and the fixed-multiplicity question (1.2) are different statistics: a theorem about $\sum_j ([x^j]P_N)^r$ does not by itself settle the number of coefficients equal to a particular integer. A further, separate question of Stanley about “Fibonacci polynomials” [2] concerns the same kind of fixed-multiplicity count for a different exponent sequence; unlike the present target, that question does have posted solutions, which establish eventual linearity of its fixed-coefficient counts,

together with indexing corrections. This report does not claim to contradict any of those results, nor to settle the other conjectures in [4].

2 From polynomial coefficients to interval coverage

The basic mechanism is that one half of the product creates a solid interval of monomials, while the other half creates a sparse set of translates. Counting coefficients equal to one then becomes a geometric counting problem on the integer line.

2.1 A general unique-coverage identity

Let $s_0 < s_1 < \dots < s_{M-1}$ be integer starting points, $M \geq 2$, and let $L \geq 1$ be an integer. Consider the half-open intervals

$$I_i = [s_i, s_i + L), \quad 0 \leq i < M.$$

Only their integer points are counted. Write $d_i = s_{i+1} - s_i$ for $0 \leq i < M - 1$.

Lemma 2.1 (Unique coverage). *The number of integers belonging to exactly one of the intervals I_i is*

$$\min(d_0, L) + \min(d_{M-2}, L) + \sum_{i=1}^{M-2} \max(0, \min(d_{i-1}, L) + \min(d_i, L) - L). \quad (2.1)$$

The sum is empty when $M = 2$.

Proof. An integer in I_0 belongs to no other interval exactly when it is smaller than s_1 . There are $\min(d_0, L)$ such integers. The last interval similarly contributes $\min(d_{M-2}, L)$.

Now fix an internal interval I_i . Because all intervals have the same length, the immediate predecessor has the rightmost endpoint among all previous intervals, and the immediate successor has the leftmost starting point among all later ones. An integer in I_i therefore belongs to no other interval exactly when it lies in

$$[\max(s_i, s_{i-1} + L), \min(s_i + L, s_{i+1})).$$

The endpoints are integers. Its number of integer points is its length when positive, and zero otherwise. Subtracting s_i from both endpoints gives

$$\max(0, \min(L, d_i) - \max(0, L - d_{i-1})),$$

which is the corresponding summand in (2.1). Each uniquely covered point belongs to exactly one interval, so these contributions are disjoint and exhaustive. $\square$

Remark 2.2 (What equal lengths are doing). The argument does *not* assume that at most two intervals overlap. Many intervals may cover an interior point, and the summands in (2.1) are frequently zero for that reason. Equality of the interval lengths is the whole content: it makes the immediate predecessor the interval with the rightmost endpoint among all earlier intervals, and the immediate successor the interval with the leftmost start among all later ones. The nearest neighbour on each side therefore detects the entire union on that side, and no other interval has to be examined.

Corollary 2.3 (Isolated large gaps). *Suppose $L \geq 2$, the first and last gaps are 1, and no two gaps greater than 1 are adjacent. If H gaps have size at least L , the number of uniquely covered integers is*

$$2 + 2H. \quad (2.2)$$

Proof. The two boundary terms in (2.1) are 1. For an internal interval at least one neighboring gap is 1. If the other gap is g , the contribution is

$$\max(0, 1 + \min(g, L) - L).$$

As g and L are integers, this is 1 exactly when $g \geq L$, and is zero otherwise. Each gap of size at least L is nonunit, is not a boundary gap, and borders two internal intervals. It therefore contributes twice. $\square$

2.2 The rectangular binary–ternary product

For $m, n \geq 0$, define

$$Q_{m,n}(x) = \prod_{j=0}^{m-1} (1 + x^{2^j}) \prod_{j=0}^{n-1} (1 + x^{3^j}). \quad (2.3)$$

The first product enumerates all binary expansions of length m , uniquely, so

$$\prod_{j=0}^{m-1} (1 + x^{2^j}) = 1 + x + \cdots + x^{2^m-1}. \quad (2.4)$$

The second product has coefficient 1 precisely at the points of

$$S_n = \left\{ \sum_{j=0}^{n-1} \epsilon_j 3^j : \epsilon_j \in \{0, 1\} \right\}, \quad |S_n| = 2^n. \quad (2.5)$$

Ternary expansions are unique, so there is no multiplicity within S_n . Consequently

$$[x^s]Q_{m,n}(x) = \#\{u \in S_n : u \leq s < u + 2^m\}. \quad (2.6)$$

This identity reduces the coefficient problem exactly, not approximately, to Lemma 2.1 with interval length $L = 2^m$.

3 Ternary carries and exact counts

3.1 The complete gap distribution

The elements of S_n are sorted by interpreting the binary digit strings $\epsilon_{n-1} \cdots \epsilon_0$ as ternary strings. This preserves their order: changing the highest differing digit outweighs all lower digits.

Lemma 3.1 (Ternary support gaps). *Write the elements of S_n in increasing order as $s_0 < s_1 < \cdots < s_{2^n-1}$. For $n \geq 1$ and $1 \leq i < 2^n$,*

$$s_i - s_{i-1} = g_{v_2(i)}, \quad g_t = \frac{3^t + 1}{2}, \quad (3.1)$$

where $v_2(i)$ is the exponent of 2 in i . Consequently the gap sizes are exactly the g_t with $0 \leq t < n$, and g_t occurs 2^{n-1-t} times. The first and last gaps are 1, and every pair of adjacent gaps contains a gap of size 1.

Proof. To move to the next binary digit string, turn a string of t trailing ones into zeroes and turn the next zero into a one. The corresponding change in the ternary value is

$$3^t - \sum_{j=0}^{t-1} 3^j = 3^t - \frac{3^t - 1}{2} = \frac{3^t + 1}{2} = g_t. \quad (3.2)$$

Passing from index $i - 1$ to index i turns exactly $t = v_2(i)$ trailing ones into zeroes, since $i - 1$ ends in t ones precisely when i ends in t zeroes. This proves (3.1). The number of indices $1 \leq i < 2^n$ with $v_2(i) = t$ is 2^{n-1-t} , which gives the multiplicity; equivalently, for a carry of length t the t trailing digits and the next digit are fixed while the $n - 1 - t$ higher digits are arbitrary. Finally, $v_2(i) = 0$ whenever i is odd, so at least one of any two consecutive indices contributes a unit gap, and $i = 1$ and $i = 2^n - 1$ are both odd. These observations establish the adjacency and boundary claims. $\square$

The indexed form (3.1) is a ruler-sequence statement: the gap at position i is determined by the 2-adic valuation of i alone. It is slightly stronger than the multiset description, it makes the adjacency claim immediate, and it is the form the verification code checks at every single index rather than in aggregate. Section 9 records the base- a generalization, in which v_2 becomes the number of trailing zeroes in base a .

The multiplicities sum to $2^n - 1$, as required. Although the largest gap grows exponentially, gaps larger than one remain separated by unit gaps. That separation is what makes the number of coefficients equal to one simple.

Example 3.2 (A visible ruler). At $n = 3$ the sorted support is

$$0, 1, 3, 4, 9, 10, 12, 13,$$

whose successive gaps are

$$1, 2, 1, 5, 1, 2, 1 \quad (= g_0, g_1, g_0, g_2, g_0, g_1, g_0).$$

Alternate gaps are one, exactly as (3.1) predicts from $v_2(1), \dots, v_2(7) = 0, 1, 0, 2, 0, 1, 0$.

3.2 An exact integer threshold

Define

$$T(m) = \min\{t \geq 0 : 3^t \geq 2^{m+1} - 1\}. \quad (3.3)$$

For $m \geq 1$, this is at least 1. Let $U(m, n)$ denote the number of coefficients equal to one in $Q_{m,n}$.

Theorem 3.3 (Rectangular count). For $m, n \geq 1$,

$$U(m, n) = 2^{1+\max(0, n-T(m))}. \quad (3.4)$$

The boundary cases are

$$U(m, 0) = 2^m, \quad U(0, n) = 2^n. \quad (3.5)$$

In particular $U(0, 0) = 1$.

Proof. Corollary 2.3 with interval length $L = 2^m$ and Lemma 3.1 give the count in its uncollapsed form

$$U(m, n) = 2 + 2 \sum_{t=0}^{n-1} 2^{n-t-1} \mathbf{1}_{\{g_t \geq 2^m\}}. \quad (3.6)$$

A gap g_t is at least 2^m exactly when $t \geq T(m)$, because the g_t increase strictly. If $T(m) < n$, the surviving terms form a geometric sum,

$$H = \sum_{t=T(m)}^{n-1} 2^{n-1-t} = 2^{n-T(m)} - 1.$$

If $T(m) \geq n$, then $H = 0$. In either case $2 + 2H$ is (3.4). When one of the products is empty, the other has all its nonzero coefficients equal to one, proving (3.5). $\square$

Display (3.6) is kept because it, rather than the collapsed formula, is what the interval-sweep verifier evaluates term by term; having it in the text ties the code directly to the theorem.

Since $P_{2n} = Q_{n,n}$ and $P_{2n+1} = Q_{n+1,n}$, this already determines the entire sequence $v(N)$ by integer arithmetic. For $n \geq 1$, the inequality

$$3^n + 1 \geq 2^{n+1} \tag{3.7}$$

holds by induction, so $T(n) \leq n$. Thus

$$a_0 = 1, \quad a_n = 2^{n-T(n)+1} \quad (n \geq 1). \tag{3.8}$$

The first values are given in Table 1, together with the odd-prefix counts derived in Section 6.

Inequality (3.7) is the sharp form of a bound that holds far more generally. Since $g_{t+1} = 3g_t - 1$ and $g_t \geq 1$,

$$g_{t+1} \geq 2g_t, \quad \text{hence} \quad g_t \geq 2^t \quad (t \geq 0). \tag{3.9}$$

Because $T(n)$ is the least t with $g_t \geq 2^n$, this alone forces $T(n) \leq n$, and $T(n) \geq 1$ for $n \geq 1$. The advantage of (3.9) is that it survives the replacement of 3 by any $q \geq 3$: the general gaps obey $g_{t+1} = qg_t - (a-1)$ and $(q-2)g_t \geq a-1$ whenever $q > a \geq 2$, so $g_t \geq 2^t$ throughout the family of Section 9. Only for $a = 2$, where the interval length is 2^m , does this yield the uniform bound $T_{2,q}(n) \leq n$; for $a \geq 3$ the interval length is a^m and the bound genuinely fails, as the case $T_{3,4}(1) = 2$ in Section 9 shows. We therefore keep (3.7) as the tight statement for (2, 3) and (3.9) as the family-wide one.

Table 1: Exact counts on both parities. The convention $T(0) = 0$ is the natural one for the threshold, while $a_0 = 1$ is handled separately because the product is empty. The last column witnesses Proposition 6.2: it equals $a_{n+1}/2$ except at $n = 0, 1, 2, 3$, where it exceeds it by one.

n	$T(n)$	a_n	$v(2n+1)$	n	$T(n)$	a_n	$v(2n+1)$
0	0	1	2	11	8	16	8
1	1	2	2	12	9	16	16
2	2	2	2	13	9	32	16
3	3	2	2	14	10	32	16
4	4	2	2	15	11	32	32
5	4	4	2	16	11	64	32
6	5	4	2	17	12	64	64
7	6	4	4	18	12	128	64
8	6	8	4	19	13	128	64
9	7	8	8	20	14	128	128
10	7	16	8				

3.3 A fully visible example

Take $m = n = 5$. The interval length is 32, while the gap sizes are 1, 2, 5, 14, 41. Only the gap 41 reaches the threshold; it occurs once. Therefore $U(5, 5) = 2 + 2 = 4$.

Concretely, the support S_5 splits into a block ending at $1 + 3 + 9 + 27 = 40$ and a block beginning at 81. The first block's final interval ends just after 71, while the second block starts at 81. The four uniquely covered integers are

$$0, \quad 71, \quad 81, \quad 152. \quad (3.10)$$

The outer two come from the global boundaries, and the middle two flank the unique large gap. Direct multiplication of P_{10} gives precisely these coefficient-one positions; Figure 1 displays all coefficients, including the zero coefficients between the two blocks.

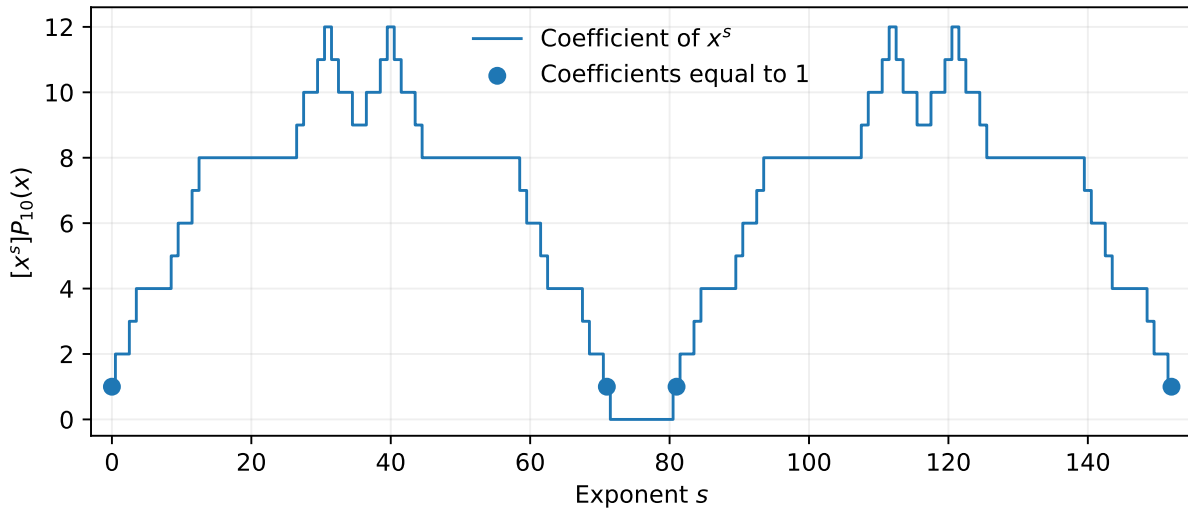


Figure 1: The coefficients of $P_{10} = Q_{5,5}$. The four marked values occur at the positions in (3.10). The maximum coefficient is 12; the sum of all coefficients is $2^{10} = 1024$.

4 An elementary obstruction to rationality

The exact count is a power of two. Its exponent grows at an irrational average speed but changes only by zero or one at each step. Rational generating functions cannot exhibit this particular combination.

4.1 Rational recurrences modulo a prime

A sequence is called *eventually C-finite* when, from some index onward, it satisfies a nontrivial linear recurrence with constant coefficients; Definition 5.1 states this and the polynomial-coefficient analogue precisely, and both notions are used again in Sections 5 and 8. An ordinary generating function over $\mathbb{Q}$ is rational if and only if its coefficients are eventually C-finite over $\mathbb{Q}$: multiplying the series by its denominator gives the recurrence, and summing an eventual recurrence gives a polynomial numerator divided by a polynomial denominator.

There is no change in this conclusion if rationality is initially allowed over $\mathbb{C}$ for a series with rational coefficients. For a fixed denominator degree and a fixed numerator bound, the coefficient identities

are linear equations with rational entries in finitely many unknown denominator coefficients, with the constant denominator coefficient normalized to one. A finite subsystem spans all the augmented rows. If a complex solution exists, rational Gaussian elimination gives a rational solution to that subsystem and hence to all the identities. Thus the rational function can be taken over $\mathbb{Q}$.

Lemma 4.1 (Finite ratios and irrational exponent slope). *Let $b \geq 2$ and $C \geq 1$ be integers. Suppose an integer sequence (u_n) has, for all sufficiently large n , the form*

$$u_n = Cb^{e_n}, \quad e_n \in \mathbb{Z}_{\geq 0}, \quad e_{n+1} - e_n \in \{0, 1\}.$$

If $\lim_{n \rightarrow \infty} e_n/n$ exists and is irrational, then $\sum_{n \geq 0} u_n z^n$ is not rational.

Proof. Suppose the generating function were rational. Then for some $d \geq 1$ and rational constants $c_0, \dots, c_{d-1}$,

$$u_{n+d} = c_{d-1}u_{n+d-1} + \dots + c_0u_n \tag{4.1}$$

for every sufficiently large n . Choose a prime p dividing neither $Cb(b-1)$ nor any denominator of the c_j . Only finitely many primes are excluded.

Reduce (4.1) modulo p . The state vector $(u_n, \dots, u_{n+d-1}) \bmod p$ belongs to the finite set $\mathbb{F}_p^d$ and has a deterministic successor. Its forward orbit, and hence $u_n \bmod p$, is eventually periodic. No invertibility of the transition is needed.

For large n , all $u_n \bmod p$ are nonzero. The quotient $u_{n+1}/u_n \bmod p$ is therefore eventually periodic as well. Its actual integer value is either 1 or b , and these have distinct residues modulo p . The periodic residue quotient thus uniquely determines the actual quotient. Consequently $e_{n+1} - e_n$ is eventually periodic.

If its eventual period has length L and contains J ones, summing the increments yields $e_n = (J/L)n + O(1)$. The limit e_n/n must be the rational number J/L , contrary to the hypothesis. $\square$

4.2 Applying the obstruction

By its definition, $T(n)$ is nondecreasing. For $n \geq 1$, if $3^{T(n)} \geq 2^{n+1} - 1$, then

$$3^{T(n)+1} \geq 3(2^{n+1} - 1) \geq 2^{n+2} - 1.$$

Therefore

$$T(n+1) - T(n) \in \{0, 1\} \quad (n \geq 1). \tag{4.2}$$

Moreover, the two adjacent powers bracketing the threshold give

$$T(n) = n \log_3 2 + O(1). \tag{4.3}$$

For $n \geq 1$ put $e_n = n - T(n) + 1$. Equations (3.8), (4.2), and (4.3) show that

$$a_n = 2^{e_n}, \quad e_{n+1} - e_n \in \{0, 1\}, \quad \frac{e_n}{n} \longrightarrow 1 - \log_3 2.$$

The last limit is irrational. Indeed, a rational equality $\log_3 2 = p/q$ with positive integers p, q would imply $2^q = 3^p$, contradicting unique prime factorization. Lemma 4.1 proves that A is not rational.

Finally, the even section of a rational series is rational. One way to see this is to represent an eventual recurrence by a finite companion matrix: sampling at even indices replaces the matrix by its square and again gives a constant-coefficient recurrence, by the characteristic polynomial. Equivalently,

$$\frac{F(t) + F(-t)}{2} = A(t^2),$$

and an even rational function of t is a rational function of t^2 . Thus rationality of F would imply rationality of A , a contradiction. This proves the negative answer to the universal rationality question.

This companion-matrix argument is exactly what the present section needs, and no more: it handles the constant-coefficient case at $d = 2$, $r = 0$. Lemma 5.4 below proves the same closure property for all arithmetic subsequences and for polynomial-coefficient recurrences, which is what the non-D-finiteness of F will require.

Remark 4.2 (What the argument does not assume). The irrationality of an exponential growth constant by itself would not rule out rationality: coefficients of a rational function can have irrational algebraic exponential growth. The proof instead uses an irrational *frequency of discrete exponent increments* and finite-state arithmetic modulo a prime. It makes no algebraicity or transcendence claim about $2^{1-\log_3 2}$.

5 Finite-quotient rigidity

The previous section is complete as an answer to the posed question. This section gives a second, independent algebraic route to the same obstruction, and the two are worth keeping side by side because they buy different things.

The modular argument of Section 4 is elementary and entirely finite: it needs no invertibility of the transition, no field extension, and no analysis. What it yields is nonrationality only, because reducing a polynomial-coefficient recurrence modulo a prime does not produce a finite-state system — the coefficients then vary with n . The rigidity theorem below works over $\mathbb{C}$, needs no prime and no reduction, and reaches the strictly stronger conclusion that the counts are not P-recursive, hence that their generating function is not D-finite. Neither argument subsumes the other in difficulty or in hypotheses, so both are given in full.

This section also supplies no recurrence guessing from data and invokes no theorem on zeros of recurrence sequences.

Definition 5.1 (Recurrence classes). A sequence (u_n) of complex numbers is *eventually C-finite* if there exist constants $c_0, \dots, c_s$, not all zero, with

$$\sum_{j=0}^s c_j u_{n+j} = 0 \quad \text{for all sufficiently large } n.$$

It is *P-recursive* if the same holds with polynomials $p_j(n) \in \mathbb{C}[n]$, not all zero, in place of the constants. Changing finitely many terms of a sequence changes neither property, since both are required only eventually.

5.1 The rigidity theorem

Theorem 5.2 (Finite-quotient rigidity). *Let (u_n) be an eventually nonzero complex sequence whose adjacent quotients u_{n+1}/u_n lie in a fixed finite set. The following are equivalent:*

- (i) (u_n) is P-recursive;
- (ii) (u_n) is eventually C-finite;
- (iii) the quotient sequence (u_{n+1}/u_n) is eventually periodic.

Proof. Discard an initial segment, so that every term used below is nonzero. Implication (ii) $\Rightarrow$ (i) is immediate from Definition 5.1, since constants are polynomials. We prove (i) $\Rightarrow$ (ii), (ii) $\Rightarrow$ (iii) and (iii) $\Rightarrow$ (ii).

(i) $\Rightarrow$ (ii). Suppose

$$\sum_{j=0}^s p_j(n)u_{n+j} = 0 \quad (5.1)$$

for all large n , with some p_j nonzero. The normalized vector

$$V_n = \left(1, \frac{u_{n+1}}{u_n}, \dots, \frac{u_{n+s}}{u_n}\right)$$

takes only finitely many values: each coordinate is a product of at most s members of the fixed finite quotient set. For each attainable vector $v = (v_0, \dots, v_s)$ form the polynomial

$$q_v(X) = \sum_{j=0}^s p_j(X)v_j.$$

If q_v is not identically zero it has only finitely many integer roots. As there are finitely many such v , there is an N_0 exceeding every nonnegative integer root of every nonzero q_v .

For $n \geq N_0$, dividing (5.1) by u_n says exactly $q_{V_n}(n) = 0$; by the choice of N_0 the polynomial q_{V_n} must therefore vanish identically. Let $d = \max_j \deg p_j$ and let c_j be the coefficient of X^d in p_j , so that at least one c_j is nonzero. Comparing coefficients of X^d in $q_{V_n} \equiv 0$ gives

$$\sum_{j=0}^s c_j \frac{u_{n+j}}{u_n} = 0, \quad \text{hence} \quad \sum_{j=0}^s c_j u_{n+j} = 0$$

for every $n \geq N_0$. This is a nontrivial constant-coefficient recurrence.

(ii) $\Rightarrow$ (iii). Choose an eventual constant-coefficient recurrence whose largest index s has $c_s \neq 0$. Necessarily $s \geq 1$, since the terms are nonzero. If $s = 1$ the quotient is eventually the constant $-c_0/c_1$. For $s \geq 2$ consider

$$W_n = \left(1, \frac{u_{n+1}}{u_n}, \dots, \frac{u_{n+s-1}}{u_n}\right),$$

which again ranges over a finite set. The recurrence determines u_{n+s}/u_n from W_n , and dividing every shifted coordinate by the nonzero entry u_{n+1}/u_n then determines W_{n+1} from W_n . So the states follow a deterministic map on a finite set; their trajectory is eventually periodic, and the second coordinate of W_n is the adjacent quotient.

(iii) $\Rightarrow$ (ii). If the quotients have eventual period p , let C be the product of the quotients over one period. Then $u_{n+p} = Cu_n$ for all large n , a constant-coefficient recurrence. $\square$

Corollary 5.3 (Irrational slope defeats P-recursiveness). *Let $c \geq 1$ be an integer, let $u_n > 0$, and suppose that $u_{n+1}/u_n \in \{1, c\}$ for all large n , with $c \geq 2$. If*

$$\lim_{n \rightarrow \infty} \frac{\log_c u_n}{n}$$

exists and is irrational, then (u_n) is not P-recursive, and therefore $\sum_n u_n z^n$ is neither rational nor D-finite.

Proof. The quotients lie in the fixed finite set $\{1, c\}$. If they were eventually periodic with period p containing exactly j occurrences of c , then $\log_c u_n = (j/p)n + O(1)$ and the displayed limit would be the rational number j/p . Theorem 5.2 therefore excludes P-recursiveness. The final clause is the standard equivalence between D-finite series and P-recursive coefficient sequences [5, Theorem 1.5], whose mechanism is recalled at the end of Section 8. $\square$

Applied with $c = 2$ to the binary–ternary counts, Corollary 5.3 already gives more than Section 4 did. Equation (3.8) together with (4.2) shows that $a_{n+1}/a_n \in \{1, 2\}$ for $n \geq 1$, and (4.3) gives

$$\frac{\log_2 a_n}{n} \longrightarrow 1 - \alpha = 1 - \log_3 2,$$

which is irrational. Hence (a_n) is not P-recursive. In Section 9 the same corollary is applied with $c = a$ to the whole two-base family, using the quotient set $\{1, a\}$ supplied by (9.13).

5.2 Passing to arithmetic subsequences

The transfer from the even section to the full sequence must be proved, not assumed; and for the D-finite conclusion it must be proved in the polynomial-coefficient setting, which the companion-matrix argument of Section 4 does not cover.

Lemma 5.4 (Decimation). *Fix integers $d \geq 1$ and $r \geq 0$. If $(u_N)_{N \geq 0}$ is P-recursive, then $(u_{dn+r})_{n \geq 0}$ is P-recursive. If (u_N) is eventually C-finite, then (u_{dn+r}) is eventually C-finite.*

Proof. An order-zero recurrence $p_0(N)u_N = 0$ with $p_0 \neq 0$ forces $u_N = 0$ for all large N , and the assertion is immediate. Otherwise take an order- s recurrence with $s \geq 1$. Away from the finitely many integer zeros of its leading coefficient it can be solved for u_{N+s} and written as

$$X_{N+1} = M(N)X_N, \quad X_N = (u_N, u_{N+1}, \dots, u_{N+s-1})^T,$$

with the entries of $M(N)$ rational functions of N . Composing d consecutive steps and sampling at $N = dn + r$ gives a first-order system

$$Y_{n+1} = B(n)Y_n, \quad Y_n = X_{dn+r},$$

of the same dimension s , with entries rational in n . The first coordinates of $Y_n, Y_{n+1}, \dots, Y_{n+s}$ are $s + 1$ linear forms in the s coordinates of Y_n , with coefficients in $\mathbb{C}(n)$; any $s + 1$ linear forms in s variables are linearly dependent over that field. Clearing denominators turns such a dependence into a nonzero polynomial-coefficient recurrence for (u_{dn+r}) . If the original coefficients are constant, then M is constant, $B = M^d$ is constant, and the same argument produces constant coefficients. $\square$

Consequently $(v(N))$ is not P-recursive either: its even subsequence (a_n) is not, and Lemma 5.4 with $d = 2, r = 0$ would otherwise make it so. This completes the second algebraic route, and it establishes non-P-recursiveness of both sequences — a conclusion that Section 4 does not reach.

6 The floor formula and the odd indices

6.1 Removing the near-integer ambiguity

The threshold (3.3) involves $2^{n+1} - 1$, not 2^{n+1} . The difference is important for the initial terms and must not be silently dropped.

Proposition 6.1 (Exact floor expression). *For $n \geq 2$,*

$$T(n) = \lceil (n+1)\alpha \rceil = \lfloor (n+1)\alpha \rfloor + 1, \quad \alpha = \log_3 2. \quad (6.1)$$

Consequently (1.7) holds for every $n \geq 0$.

Proof. Let $M = \lceil (n+1)\alpha \rceil$ be the least integer such that $3^M \geq 2^{n+1}$. The number $(n+1)\alpha$ is irrational, so the inequality is strict and the ceiling is one more than the floor. Since the smaller threshold in (3.3) is $2^{n+1} - 1$, we have $T(n) \leq M$. If $T(n) < M$, an integer power of three is at least $2^{n+1} - 1$ but smaller than 2^{n+1} . It must equal $2^{n+1} - 1$. For $n \geq 2$, the latter is 7 modulo 8, whereas a power of three is 1 or 3 modulo 8. This is impossible.

Substitute (6.1) into (3.8). At $n = 0$, $2^{0-\lfloor \alpha \rfloor} = 1 = a_0$. At $n = 1$, $2^{1-\lfloor 2\alpha \rfloor} = 1$ while $a_1 = 2$, producing exactly the correction $\mathbf{1}_{\{n=1\}}$. $\square$

For $n \geq 2$ the logarithmic exponent increments can also be written as

$$\log_2 \frac{a_{n+1}}{a_n} = 1 - (\lfloor (n+2)\alpha \rfloor - \lfloor (n+1)\alpha \rfloor). \quad (6.2)$$

This is a binary sequence driven by an irrational rotation. Its long-run frequency of ones is $1 - \alpha$, either by telescoping the floors or by the equidistribution argument in the next section. In particular, the proportion of indices at which a_n doubles tends to $1 - \alpha$.

6.2 The doubling word

It is worth making the coding in (6.2) completely explicit, because it is the exact point at which an irrational number enters a purely combinatorial sequence. Write $u_n = \{(n+1)\alpha\}$, as in (7.1) below. For $n \geq 2$ the value (6.2) equals 1 — that is, $a_{n+1} = 2a_n$ — precisely when $\lfloor (n+2)\alpha \rfloor = \lfloor (n+1)\alpha \rfloor$, and hence precisely when

$$u_n < 1 - \alpha. \quad (6.3)$$

Otherwise $a_{n+1} = a_n$. The doubling word $(\mathbf{1}_{\{u_n < 1-\alpha\}})_{n \geq 2}$ is therefore a Sturmian coding of the rotation by α , and by Lemma 7.1 its limiting proportion of ones is the length $1 - \alpha$ of the coding interval.

This yields the obstruction of Section 5 in one line: no eventually periodic binary word has an irrational proportion of ones, because the proportion of an eventually periodic word is the number of ones in one period divided by the period length. So the doubling word is not eventually periodic, so by Theorem 5.2 the sequence (a_n) — whose adjacent quotients are exactly the letters of that word, read as 1 and 2 — is not P-recursive.

6.3 The entire product-length sequence

The rectangular formula also determines the odd indices without a new counting argument.

Proposition 6.2 (Even–odd identity). *For every $n \geq 0$,*

$$b_n = \frac{a_{n+1}}{2} + \mathbf{1}_{\{0 \leq n \leq 3\}}. \quad (6.4)$$

Thus

$$F(t) = \frac{(2t+1)A(t^2) - 1}{2t} + t + t^3 + t^5 + t^7, \quad (6.5)$$

where the apparent singularity at $t = 0$ is removable.

Proof. For $n \geq 1$, Theorem 3.3 gives $b_n = 2^{1+\max(0, n-T(n+1))}$, while $a_{n+1}/2 = 2^{n+1-T(n+1)}$. The bound $T(n+1) \leq n+1$ shows that these agree unless $T(n+1) = n+1$, in which case their values are 2 and 1. Directly, $T(2) = 2$, $T(3) = 3$, and $T(4) = 4$. For $n \geq 4$ the inequality $3^n \geq 2^{n+2} - 1$ follows from its case $n = 4$ by induction; hence $T(n+1) \leq n$. At $n = 0$ the required identity is $b_0 = 2 = a_1/2 + 1$.

Writing $B(z) = \sum_{n \geq 0} b_n z^n$, summation gives

$$B(z) = \frac{A(z) - 1}{2z} + 1 + z + z^2 + z^3.$$

Substitute this into $F(t) = A(t^2) + tB(t^2)$ to obtain (6.5). $\square$

Proposition 6.2 is the $(a, q) = (2, 3)$ instance of a general even–odd identity; Proposition 9.3 and Corollary 9.4 in Section 9 give the statement for every pair of bases, with the exceptional index set computed explicitly for $a = 2$. The correction $\mathbf{1}_{\{0 \leq n \leq 3\}}$ here is not an artefact of small numbers but the complete list of indices at which the threshold saturates; for $q \geq 5$ the analogous list is empty.

The initial values, indexed by the actual number of factors, are

$$(\nu(N))_{N=0}^{22} = (1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 4, 2, 4, 2, 4, 4, 4, 8, 4, 8, 8, 16, 8, 16). \quad (6.6)$$

The drops at odd indices are real; adding a factor can turn many coefficients that were one into larger coefficients. The even subsequence is monotone, but the full coefficient-one sequence need not be.

7 Asymptotics with persistent oscillation

Set

$$\lambda = 2^{1-\alpha}, \quad R = \lambda^{-1}, \quad h(u) = 2^{u-\alpha} \quad (0 \leq u < 1), \quad u_n = \{(n+1)\alpha\}. \quad (7.1)$$

Numerically,

$$\alpha \approx 0.630929753571, \quad \lambda \approx 1.291520234330, \quad R \approx 0.774281326315.$$

These decimal approximations play no role in the exact computations or proofs. Proposition 6.1 gives the exact representation

$$a_n = \lambda^n h(u_n) + \mathbf{1}_{\{n=1\}}. \quad (7.2)$$

Thus the relative fluctuations do not shrink with n ; they are the values of a fixed function sampled along an irrational rotation.

7.1 The equidistribution input

We use the following elementary consequence of the geometric-sum formula.

Lemma 7.1 (Irrational rotation averages). *If β is irrational and f is Riemann integrable on $[0, 1]$, with its values interpreted periodically, then for every real γ ,*

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=0}^{M-1} f(\{n\beta + \gamma\}) = \int_0^1 f(u) du. \quad (7.3)$$

Proof. For each nonzero integer j ,

$$\frac{1}{M} \sum_{n=0}^{M-1} e^{2\pi i j(n\beta + \gamma)} = \frac{e^{2\pi i j \gamma}}{M} \frac{1 - e^{2\pi i j M \beta}}{1 - e^{2\pi i j \beta}} \longrightarrow 0.$$

For $j = 0$ the average is 1. Therefore (7.3) holds for trigonometric polynomials. Uniform approximation by trigonometric polynomials extends it to continuous periodic functions. Continuous upper and lower approximations to an interval indicator then give its average as the interval length: the approximations can differ only in arbitrarily short neighborhoods of the endpoints. Finite step functions follow. Finally, upper and lower Riemann sums approximate every real Riemann integrable f with arbitrarily small integral gap. Apply the same argument to the real and imaginary parts for complex-valued f . $\square$

Proposition 7.2 (Growth and fluctuation law). *The sequence a_n satisfies*

$$\lim_{n \rightarrow \infty} a_n^{1/n} = \lambda, \quad a_n = \Theta(\lambda^n). \quad (7.4)$$

The set of limit points of a_n/λ^n is the whole interval

$$[2^{-\alpha}, 2^{1-\alpha}]. \quad (7.5)$$

In particular there is no constant C for which $a_n \sim C\lambda^n$. Its normalized Cesàro mean does exist:

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=0}^{M-1} \frac{a_n}{\lambda^n} = \frac{2^{-\alpha}}{\log 2}. \quad (7.6)$$

The full sequence has root growth $\lim_{N \rightarrow \infty} v(N)^{1/N} = \sqrt{\lambda}$.

Proof. Outside the single corrected index, (7.2) lies between the two fixed positive bounds $2^{-\alpha}\lambda^n$ and $2^{1-\alpha}\lambda^n$. This proves (7.4). Lemma 7.1 implies that the fractional parts u_n visit every subinterval of $[0, 1)$ infinitely often. The strictly increasing function h therefore gives every interior point of (7.5) as a subsequential limit; approaching 0 and 1 gives the endpoints. No other limits are possible. Equation (7.6) follows by integrating h , and the finite exceptional term vanishes in the average. Finally, (6.4) gives the same root growth on both parities of $v(N)$. $\square$

7.2 The exact law of the fluctuations

Proposition 7.2 identifies the closure of the set of fluctuation values and their mean. Equidistribution in fact delivers the entire limiting distribution, not merely its support and its first moment.

Proposition 7.3 (Limiting distribution of the normalized counts). *For every y with $2^{-\alpha} \leq y \leq 2^{1-\alpha}$,*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \# \left\{ 2 \leq n < N : \frac{a_n}{\lambda^n} \leq y \right\} = \alpha + \log_2 y. \quad (7.7)$$

The limiting law is supported on $[2^{-\alpha}, 2^{1-\alpha}]$ with density $1/(y \log 2)$ there, and its mean is the Cesàro value $2^{-\alpha}/\log 2$ of (7.6).

Proof. For $n \geq 2$, (7.2) gives $a_n/\lambda^n = h(u_n) = 2^{u_n - \alpha}$ with $u_n = \{(n+1)\alpha\}$. Since h is strictly increasing, the event $h(u_n) \leq y$ is the event $u_n \leq \alpha + \log_2 y$, and the right-hand side lies in $[0, 1]$ exactly on the stated range of y . Apply Lemma 7.1 to the indicator of $[0, \alpha + \log_2 y]$, which is Riemann integrable, with $\beta = \alpha$ and $\gamma = \alpha$. The density is obtained by differentiating $\alpha + \log_2 y$, and $\int_{2^{-\alpha}}^{2^{1-\alpha}} y \cdot \frac{dy}{y \log 2} = \int_0^1 2^{u-\alpha} du = 2^{-\alpha}/\log 2$. $\square$

Remark 7.4 (Two normalizations of the same law). It is equally natural to divide by λ^{n+1} rather than λ^n . Because $\lambda = 2^{1-\alpha}$ and, by irrationality, $\{(n+1)(1-\alpha)\} = 1 - \{(n+1)\alpha\}$, the two normalized quantities are related by

$$\frac{a_n}{\lambda^{n+1}} = \frac{1}{\lambda} \cdot \frac{a_n}{\lambda^n} = 2^{u_n-1} = 2^{-\{(n+1)(1-\alpha)\}} \quad (n \geq 2). \quad (7.8)$$

In that normalization the support is $[1/2, 1]$, the distribution function (7.7) reads $1 + \log_2 y$, and the mean becomes $\int_0^1 2^{-x} dx = 1/(2 \log 2)$. The two means differ by exactly the factor λ , as (7.8) requires; neither is a competing claim about the same quantity. All statements in this report use the λ^n normalization of (7.1).

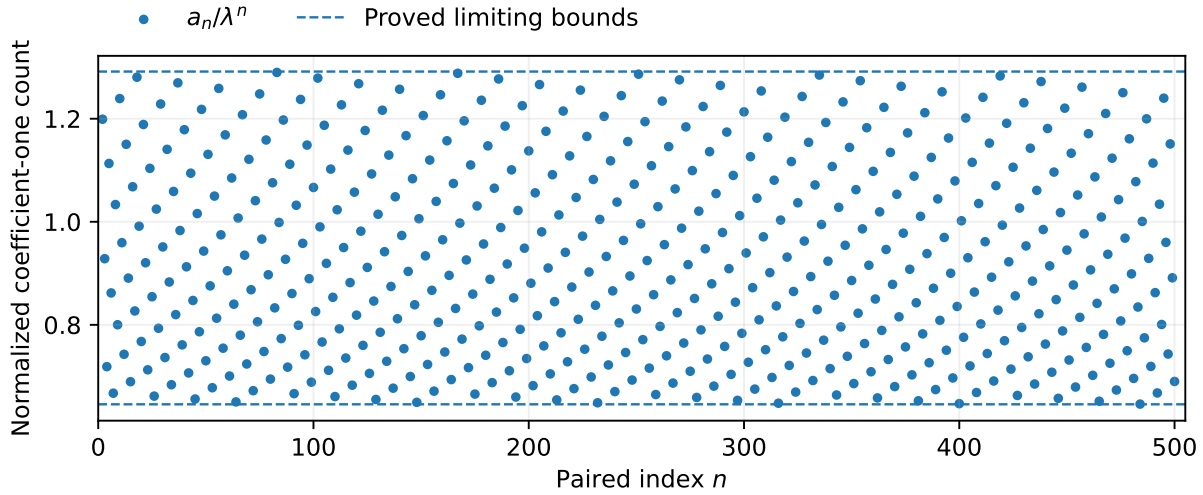


Figure 2: The normalized counts a_n/λ^n for $2 \leq n \leq 500$. The dashed lines are the limiting bounds in (7.5). The nonvanishing oscillation is proved by irrational rotation, not inferred from this plot. Floating-point arithmetic is used only to draw the figure.

8 Dense singularities and a natural boundary

The fluctuation law does more than prevent a constant leading asymptotic. Every Fourier mode of the profile h is nonzero. Each mode produces a boundary singularity, and the corresponding points are dense on the circle of convergence.

8.1 A precise Abelian lemma

Lemma 8.1 (Cesàro means imply radial limits). *Let (c_n) be a bounded complex sequence. If $M^{-1} \sum_{n=0}^{M-1} c_n \rightarrow c$, then*

$$\lim_{\rho \uparrow 1} (1 - \rho) \sum_{n \geq 0} c_n \rho^n = c. \quad (8.1)$$

Proof. Put $S_N = \sum_{n=0}^N c_n$. Absolute convergence for $0 < \rho < 1$ gives

$$\sum_{n \geq 0} c_n \rho^n = (1 - \rho) \sum_{N \geq 0} S_N \rho^N.$$

The hypothesis is $S_N = (N + 1)c + o(N + 1)$. Since $\sum_{N \geq 0} (N + 1)\rho^N = (1 - \rho)^{-2}$, the principal term contributes c after multiplying by $1 - \rho$. For the error, fix $\varepsilon > 0$ and choose N_0 so that its absolute value is at most $\varepsilon(N + 1)$ for $N \geq N_0$. The tail contributes at most ε , and the finitely many initial terms tend to zero. This proves (8.1). $\square$

8.2 Nonzero radial amplitudes at a dense set

Theorem 8.2 (Explicit dense boundary singularities). *For each $k \in \mathbb{Z}$, put $\omega_k = e^{-2\pi i k \alpha}$. Then*

$$\lim_{\rho \uparrow 1} (1 - \rho)A(R\rho\omega_k) = \frac{2^{-\alpha} e^{2\pi i k \alpha}}{\log 2 - 2\pi i k} \neq 0. \quad (8.2)$$

The circle $|z| = R$ is a natural boundary of A .

Proof. Equation (7.4) shows that the radius of convergence is exactly R . Ignore the single correction at $n = 1$ for the moment. At $z = R\rho\omega_k$, formula (7.2) makes the summand $h(u_n)\omega_k^n \rho^n$. Since

$$\omega_k^n = e^{2\pi i k \alpha} e^{-2\pi i k u_n},$$

Lemma 7.1, applied to the Riemann integrable function $h(u)e^{-2\pi i k u}$, gives the Cesàro mean

$$\begin{aligned} \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=0}^{M-1} h(u_n)\omega_k^n &= e^{2\pi i k \alpha} \int_0^1 2^{u-\alpha} e^{-2\pi i k u} du \\ &= \frac{2^{-\alpha} e^{2\pi i k \alpha}}{\log 2 - 2\pi i k}. \end{aligned}$$

In the integral, $e^{\log 2 - 2\pi i k} - 1 = 1$. The resulting value is nonzero for every integer k . Lemma 8.1 proves (8.2). The omitted correction contributes $(1 - \rho)R\rho\omega_k$, which tends to zero.

If A admitted a holomorphic extension across the point $R\omega_k$, it would remain bounded along the inward radial segment near that point. Equation (8.2) contradicts this. Thus every $R\omega_k$ is a singular point. Irrationality of α makes the set of ω_k dense on the unit circle. An extension across any other point of $|z| = R$ would be defined in an open neighborhood containing some $R\omega_k$, again a contradiction. Hence no point on the circle admits analytic continuation through the circle. $\square$

The limit (8.2) describes a radial blowup. It does *not* assert that these are isolated poles or provide residues of a meromorphic continuation: the singularities are dense, and no such neighborhood of continuation is available.

8.3 A second proof, from a general irrational-floor theorem

Theorem 8.2 is a direct computation on the actual coefficient profile h of (7.1). The same conclusion follows from a general principle in which the profile is first normalized to the clean exponential q^{-x} , at the cost of an extra transfer step. Both proofs are kept: the first stays inside the concrete problem and needs no auxiliary series, while the second proves considerably more — it covers every irrational exponent and every base at once, so it settles the whole family of Section 9 without a new computation.

Let $0 < \theta < 1$ be irrational and let $q > 1$ be real. Define

$$\Phi_{\theta,q}(w) = \sum_{n \geq 0} q^{\lfloor n\theta \rfloor} w^n, \quad \Lambda = q^\theta, \quad (8.3)$$

and its normalization

$$\Psi(w) = \Phi_{\theta,q}(w/\Lambda) = \sum_{n \geq 0} q^{-\{n\theta\}} w^n. \quad (8.4)$$

The coefficients of Ψ lie between q^{-1} and 1, so its radius of convergence is exactly 1.

Theorem 8.3 (Natural boundary for an irrational-floor series). *For every irrational $\theta \in (0, 1)$ and every $q > 1$, the circle $|w| = q^{-\theta}$ is a natural boundary of $\Phi_{\theta,q}$. More precisely, for each $k \in \mathbb{Z}$, with $\zeta_k = e^{-2\pi i k \theta}$,*

$$\lim_{\rho \uparrow 1} (1 - \rho) \Psi(\rho \zeta_k) = \frac{1 - q^{-1}}{\log q + 2\pi i k} \neq 0, \quad (8.5)$$

and the unit circle is a natural boundary of Ψ .

Proof. On $0 \leq x < 1$ put $f(x) = q^{-x}$ and extend f periodically. For a fixed integer k the function $f(x)e^{-2\pi i k x}$ is bounded and Riemann integrable. Since $\zeta_k^n = e^{-2\pi i k n \theta} = e^{-2\pi i k \{n\theta\}}$, the n th coefficient of Ψ evaluated along the ray through ζ_k is $f(\{n\theta\})e^{-2\pi i k \{n\theta\}}$. Lemma 7.1 with $\beta = \theta$ and $\gamma = 0$ therefore gives the Cesàro mean

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} q^{-\{n\theta\}} \zeta_k^n = \int_0^1 q^{-x} e^{-2\pi i k x} dx = \frac{1 - q^{-1}}{\log q + 2\pi i k},$$

using $e^{-\log q - 2\pi i k} = q^{-1}$. The value is nonzero for every k because $q > 1$. Lemma 8.1, applied to the bounded sequence $q^{-\{n\theta\}} \zeta_k^n$, converts this into (8.5).

If Ψ extended holomorphically through ζ_k it would be bounded on a short inward radial segment ending there, forcing the limit in (8.5) to vanish. So every ζ_k is a singular point of Ψ . Irrationality of θ makes $\{\zeta_k : k \in \mathbb{Z}\}$ dense on the unit circle, and an extension through any boundary point would be defined on an open arc, which contains some ζ_k . Hence the unit circle is a natural boundary of Ψ , and the substitution $w \mapsto w/\Lambda$ transports this to the circle $|w| = \Lambda^{-1} = q^{-\theta}$ for $\Phi_{\theta,q}$. $\square$

Remark 8.4 (Averages, not formal Fourier series). The proof evaluates explicit Cesàro averages of the sequence $q^{-\{n\theta\}} \zeta_k^n$. It does not expand the discontinuous periodic profile in a Fourier series and then evaluate that series termwise. The distinction matters: the periodic extension of q^{-x} has a jump at the integers, and a formal Fourier expansion would need a separate convergence argument exactly there. Lemma 7.1 is stated for Riemann integrable f precisely so that no such argument is needed. This complements, and does not repeat, the caution after (8.2): that remark concerns what the radial limit does not say about poles or residues, whereas this one concerns how the limit is computed.

Transfer to the count series

Applying Theorem 8.3 to the counts requires an exact bridge, because the floor in (1.7) is written with α and the theorem is stated with the complementary exponent. The two agree: since $(n+1)\alpha$ is irrational,

$$\lfloor (n+1)(1-\alpha) \rfloor = (n+1) - \lceil (n+1)\alpha \rceil = n - \lfloor (n+1)\alpha \rfloor \quad (n \geq 0), \quad (8.6)$$

so the exponent $n - \lfloor (n+1)\alpha \rfloor$ of (1.7) is the floor $\lfloor (n+1)\theta \rfloor$ of Theorem 8.3 at $\theta = 1 - \alpha$ and $q = 2$. This identity is stated once here and used without further comment; the two closed forms are the same closed form.

Write $\Phi = \Phi_{1-\alpha,2}$, so that $\Lambda = 2^{1-\alpha} = \lambda$ and the radius of convergence of Φ is $\lambda^{-1} = R$. Then

$$A(z) = \frac{\Phi(z) - 1}{z} + z. \quad (8.7)$$

Indeed, $[z^n](\Phi(z) - 1)/z = 2^{\lfloor (n+1)(1-\alpha) \rfloor} = 2^{n - \lfloor (n+1)\alpha \rfloor}$ by (8.6), which is a_n for every n except $n = 1$, where it is 1 instead of 2; the added z corrects exactly that term, in agreement with the $1_{\{n=1\}}$ of (1.7). Equation (8.7) is equivalent to $\Phi(z) = z(A(z) - z) + 1$, so each of A and Φ is an analytic function of the other on any open set avoiding $z = 0$. A holomorphic extension of A through a point of $|z| = R$ would therefore extend Φ through the same point, which Theorem 8.3 forbids. This reproves the last assertion of Theorem 8.2 from a statement that makes no reference to the enumeration.

Remark 8.5 (Reconciling the two radial constants). Theorem 8.2 and Theorem 8.3 display constants that look different — the sign of $2\pi i k$ is opposite and one carries a phase $e^{2\pi i k \alpha}$ — but they are the same statement in two index conventions, and (8.7) converts one into the other exactly. First, the two families of boundary directions coincide with reversed index: with $\theta = 1 - \alpha$,

$$\zeta_k = e^{-2\pi i k(1-\alpha)} = e^{2\pi i k \alpha} = \omega_{-k}, \quad \text{equivalently} \quad \omega_k = \zeta_{-k}.$$

So (8.5) at $q = 2$, read at the direction ω_k , gives $\lim(1 - \rho)\Psi(\rho\omega_k) = \frac{1}{2}/(\log 2 - 2\pi i k)$. Second, multiplying (8.7) by $1 - \rho$ at $z = R\rho\omega_k$ and letting $\rho \uparrow 1$ kills the terms $-(1 - \rho)/z$ and $(1 - \rho)z$ and leaves

$$\lim_{\rho \uparrow 1} (1 - \rho)A(R\rho\omega_k) = \frac{1}{R\omega_k} \lim_{\rho \uparrow 1} (1 - \rho)\Psi(\rho\omega_k) = \frac{e^{2\pi i k \alpha}}{2^{\alpha-1}} \cdot \frac{1/2}{\log 2 - 2\pi i k},$$

using $\Phi(R\rho\omega_k) = \Psi(\rho\omega_k)$ and $R = 2^{\alpha-1}$. As $\frac{1}{2} \cdot 2^{1-\alpha} = 2^{-\alpha}$, the right-hand side is $2^{-\alpha} e^{2\pi i k \alpha} / (\log 2 - 2\pi i k)$, which is exactly (8.2). The apparent discrepancy is entirely the index reversal $k \mapsto -k$ together with the factor $1/(R\omega_k)$ contributed by the transfer.

8.4 The full generating function and recurrence consequences

Corollary 8.6. *The circle $|t| = \sqrt{R}$ is a natural boundary of F .*

Proof. Equation (6.5) makes F holomorphic for $|t| < \sqrt{R}$, with a removable singularity at zero. Suppose it extended through some point t_0 on the boundary. Since $t_0 \neq 0$, the map $t \mapsto t^2$ has a local analytic inverse near t_0 . Also $2t_0 + 1 \neq 0$: the only zero is $-1/2$, whereas $\sqrt{R} > 1/\sqrt{2} > 1/2$. Solving (6.5) for $A(t^2)$ would therefore provide an analytic extension of A through t_0^2 , contrary to Theorem 8.2. $\square$

A power series $H(z)$ is D-finite if it satisfies a nonzero homogeneous linear differential equation

$$p_d(z)H^{(d)}(z) + \cdots + p_0(z)H(z) = 0, \quad p_j(z) \in \mathbb{C}[z], \quad p_d \neq 0. \quad (8.8)$$

This is the standard terminology developed in [5], where the equivalence between D-finite series and P-recursive coefficient sequences is [5, Theorem 1.5]; the direct mechanism is recalled below so that nothing here depends on an unstated import. Away from the finitely many zeros of p_d , a linear differential equation has analytic coefficients after division by p_d ; its local solutions continue along paths avoiding those zeros. An analytic germ satisfying (8.8) cannot therefore have a natural boundary on a finite circle.

Likewise, an algebraic germ has only finitely many possible finite singularities. For an irreducible polynomial equation $p(z, H) = 0$ of positive degree in H , exclude the zeros of its leading coefficient in H and of its discriminant in H ; the implicit-function theorem gives local analytic branches at the remaining points. This also contradicts a natural boundary.

Finally, a sequence (u_n) is P-recursive if it satisfies an eventual nonzero linear recurrence with polynomial coefficients in n . Such a recurrence implies a D-finite generating function. To recall the direct mechanism, let $\theta = z d/dz$. Then $\theta(z^n) = nz^n$, so sums of $p_j(n)u_{n+j}z^n$ can be written

as differential operators in θ applied to $H(z) = \sum u_n z^n$, up to finitely many initial terms. After clearing powers of z , the recurrence becomes a nonzero linear differential equation with a polynomial right-hand side. Differentiating sufficiently many times makes that right-hand side zero. These observations and the two natural-boundary results prove all remaining conclusions of Theorem 1.1.

9 A complete classification for two bases

The counterexample is part of a family in which the rationality question can be answered exactly. Fix integers $2 \leq a < q$ and define the digit polynomial

$$D_a(y) = 1 + y + \cdots + y^{a-1}. \quad (9.1)$$

Use the interleaved exponents

$$G_{2j+1} = a^j, \quad G_{2j+2} = q^j, \quad (9.2)$$

and the products $\prod_{i=1}^N D_a(x^{G_i})$. This is precisely (1.2) with fixed $r = a - 1$. The exponent generating function is $t/(1 - at^2) + t^2/(1 - qt^2)$, and $G_N \rightarrow \infty$. Let $v_{a,q}(N)$ count the coefficients equal to one and put

$$F_{a,q}(t) = \sum_{N \geq 0} v_{a,q}(N) t^N, \quad A_{a,q}(z) = \sum_{n \geq 0} v_{a,q}(2n) z^n, \quad (9.3)$$

so that $A_{a,q}$ is the even section of $F_{a,q}$, exactly as A is of F in (1.6). Two integers $a, q > 1$ are *multiplicatively dependent* when $q^u = a^v$ for some positive integers u, v ; equivalently, $\log_q a$ is rational.

Theorem 9.1 (Two-base classification). *For every pair of integers $2 \leq a < q$, the following five statements are equivalent:*

- (i) a and q are multiplicatively dependent;
- (ii) $A_{a,q}(z)$ is rational;
- (iii) $A_{a,q}(z)$ is D -finite;
- (iv) $F_{a,q}(t)$ is rational;
- (v) $F_{a,q}(t)$ is D -finite.

In particular

$$F_{a,q}(t) \text{ is rational} \iff a \text{ and } q \text{ are multiplicatively dependent}, \quad (9.4)$$

and when the bases are independent neither series is D -finite and neither count sequence is P -recursive. When $q^u = a^v$ with $u, v \geq 1$, necessarily $v > u$, and

$$v_{a,q}(N + 2v) = a^{v-u} v_{a,q}(N) \quad (9.5)$$

for every sufficiently large N . In particular the reduced denominator of $F_{a,q}(t)$ divides $1 - a^{v-u} t^{2v}$.

For $a = 2$ the dependence condition takes a familiar shape: 2 and q are multiplicatively dependent exactly when q is a power of two, since $q^u = 2^v$ forces every prime divisor of q to be 2. So the statement “ q is a power of two” that governs the binomial subfamily $D_2(y) = 1 + y$ is precisely the $a = 2$ instance of multiplicative dependence, and Theorem 9.1 contains it.

The proof occupies the rest of this section and closes the cycle

$$(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i), \quad (i) \Rightarrow (iv) \Rightarrow (v) \Rightarrow (iii).$$

Subsection 9.1 proves the enumerative formula that everything rests on. Subsection 9.2 proves $(iii) \Rightarrow (i)$ and $(v) \Rightarrow (iii)$, both in contrapositive form and both using Section 5; the second also uses Lemma 5.4. Subsection 9.3 proves $(i) \Rightarrow (ii)$ and $(i) \Rightarrow (iv)$. The implications $(ii) \Rightarrow (iii)$ and $(iv) \Rightarrow (v)$ are immediate, because a rational function satisfies a first-order linear differential equation with polynomial coefficients. Subsection 9.4 is not needed for the equivalence itself; it makes the passage between $A_{a,q}$ and $F_{a,q}$ explicit, and generalizes Proposition 6.2.

9.1 Exact counts for a general rectangle

Write

$$Q_{m,n}^{a,q}(x) = \prod_{j=0}^{m-1} D_a(x^{a^j}) \prod_{j=0}^{n-1} D_a(x^{q^j}), \quad (9.6)$$

and let $U_{a,q}(m, n)$ be its number of coefficients equal to one. The first product is $1 + x + \dots + x^{a^m-1}$. The second is supported, with coefficient one, on the a^n integers whose n base- q digits belong to $\{0, \dots, a-1\}$.

The sorted support is traversed by a base- a counter interpreted in base q . A carry through t trailing maximal digits produces the gap

$$g_t = q^t - (a-1) \sum_{j=0}^{t-1} q^j = \frac{(q-a)q^t + a-1}{q-1}. \quad (9.7)$$

It occurs

$$(a-1)a^{n-1-t} \quad (9.8)$$

times: there are $a-1$ choices for the incremented nonmaximal digit and a^{n-1-t} choices for the higher digits. Because $q > a$, the gap sizes are strictly increasing, beginning with $g_0 = 1$. A nonunit gap is preceded and followed by unit gaps, and the first and last gaps are unit, just as in Lemma 3.1.

The indexed form (3.1) generalizes verbatim. Writing the sorted support as $s_0 < s_1 < \dots < s_{a^n-1}$, the step from s_{i-1} to s_i carries through exactly as many trailing digits equal to $a-1$ as i has trailing zeroes in base a . Hence

$$s_i - s_{i-1} = g_{v_a(i)} \quad (1 \leq i < a^n), \quad (9.9)$$

where $v_a(i)$ is the largest t with $a^t \mid i$; the multiplicity (9.8) is the number of such i with $v_a(i) = t$. For $a = 2$ this is the 2-adic ruler sequence of Lemma 3.1. The verification code checks (9.9) index by index.

The exponent sequence (9.2) satisfies the order-four constant-coefficient recurrence

$$G_{N+4} = (a+q)G_{N+2} - aq G_N \quad (N \geq 1), \quad (9.10)$$

which is (1.5) at $(a, q) = (2, 3)$ and is the denominator $1 - (a+q)t^2 + aqt^4 = (1 - at^2)(1 - qt^2)$ of the exponent generating function read as a recurrence.

Define the exact threshold

$$T_{a,q}(m) = \min\{t \geq 0 : (q-a)q^t + a-1 \geq (q-1)a^m\} \quad (9.11)$$

$$= \left\lceil \log_q \left(\frac{(q-1)a^m - (a-1)}{q-a} \right) \right\rceil. \quad (9.12)$$

The ceiling expression is an identity over the real numbers; the integer inequality in (9.11) is the appropriate computational definition.

Proposition 9.2 (General rectangular formula). *For $m, n \geq 1$,*

$$U_{a,q}(m, n) = 2a^{\max(0, n - T_{a,q}(m))}. \quad (9.13)$$

The empty-product cases are $U_{a,q}(m, 0) = a^m$ and $U_{a,q}(0, n) = a^n$.

Proof. Use Corollary 2.3 with interval length a^m . As in (3.6), this gives the uncollapsed count

$$U_{a,q}(m, n) = 2 + 2 \sum_{t=0}^{n-1} (a-1)a^{n-t-1} \mathbf{1}_{\{g_t \geq a^m\}}. \quad (9.14)$$

If $T = T_{a,q}(m) < n$, the number of gaps of size at least a^m is

$$H = \sum_{t=T}^{n-1} (a-1)a^{n-1-t} = a^{n-T} - 1.$$

Otherwise $H = 0$. In either case $2 + 2H$ is (9.13). The boundary cases follow from unique base expansions. $\square$

For clarity, the factor 2 in (9.13) is not replaced by a : it counts the two uniquely covered endpoints associated with each large gap, together with the two outer boundary contributions.

9.2 Multiplicative independence forces nonrationality

The gaps satisfy

$$g_{t+1} = qg_t - (a-1). \quad (9.15)$$

For $m \geq 1$, if $g_t \geq a^m$, then

$$g_{t+1} \geq qa^m - (a-1) \geq a^{m+1},$$

because $(q-a)a^m \geq a-1$. Hence

$$T_{a,q}(m+1) - T_{a,q}(m) \in \{0, 1\} \quad (m \geq 1). \quad (9.16)$$

The restriction $m \geq 1$ is necessary for this assertion: for example, $T_{3,4}(0) = 0$ and $T_{3,4}(1) = 2$. From (9.12),

$$\frac{T_{a,q}(m)}{m} \longrightarrow \log_q a < 1. \quad (9.17)$$

The implied $O(1)$ can be made completely explicit, which is what the streaming computation of Section 10 needs. Put

$$c = c_{a,q} = \frac{q-a}{q-1} \in (0, 1). \quad (9.18)$$

From (9.7), $(q-a)q^t \leq (q-a)q^t + a-1$ and $(q-a)q^t + a-1 \leq (q-1)q^t$, so

$$c q^t \leq g_t \leq q^t \quad (t \geq 0). \quad (9.19)$$

The upper bound applied at $t = T_{a,q}(m)$ gives $q^T \geq g_T \geq a^m$, and the lower bound shows that $q^t \geq a^m/c$ already forces $g_t \geq a^m$. Hence

$$m \log_q a \leq T_{a,q}(m) \leq \left\lceil m \log_q a + \log_q(1/c) \right\rceil \quad (m \geq 0). \quad (9.20)$$

This is an explicit two-sided bound with an additive constant depending only on the pair of bases; it implies (9.17), and at $(a, q) = (2, 3)$, where $c = 1/2$, it sharpens the unquantified statement (4.3) to $n \log_3 2 \leq T(n) \leq \lceil n \log_3 2 + \log_3 2 \rceil$.

For $n \geq 1$, the even subsequence is

$$v_{a,q}(2n) = 2a^{e_n}, \quad e_n = \max(0, n - T_{a,q}(n)). \quad (9.21)$$

By (9.16), the increments of e_n belong to $\{0, 1\}$; by (9.17), $e_n/n \rightarrow 1 - \log_q a$. If a, q are multiplicatively independent, this limit is irrational: $\log_q a = p/p'$ with positive integers would give $q^p = a^{p'}$.

Two conclusions now follow, by the two routes of Sections 4 and 5.

First, applying Lemma 4.1 with $C = 2$ and $b = a$ shows that $A_{a,q}$ is not rational, hence neither is $F_{a,q}$. This is the elementary route and it needs no more.

Second — and this is the arm that (9.4) alone does not provide — (9.21) makes the adjacent quotients

$$\frac{v_{a,q}(2n+2)}{v_{a,q}(2n)} = a^{e_{n+1}-e_n} \in \{1, a\}$$

lie in a fixed finite set for $n \geq 1$, while $\log_a v_{a,q}(2n)/n \rightarrow 1 - \log_q a$ is irrational. Corollary 5.3 with $c = a$ therefore shows that $(v_{a,q}(2n))_n$ is not P-recursive, so $A_{a,q}$ is not D-finite. If $F_{a,q}$ were D-finite, then $(v_{a,q}(N))_N$ would be P-recursive, and Lemma 5.4 with $d = 2$, $r = 0$ would make $(v_{a,q}(2n))_n$ P-recursive as well. Hence $F_{a,q}$ is not D-finite either.

Nothing in this argument used $a = 2$. The finite quotient set is $\{1, a\}$ for every pair, supplied by (9.13) through (9.21), and Corollary 5.3 was stated for a general integer quotient $c \geq 2$ for exactly this reason. This proves (iii) $\Rightarrow$ (i) and (v) $\Rightarrow$ (iii) of Theorem 9.1 in contrapositive form.

9.3 Multiplicative dependence gives an eventual recurrence

Suppose $q^u = a^v$ for positive integers u, v . Since $a < q$, necessarily $u < v$. Put

$$M = a^v = q^u, \quad K = \frac{q-1}{q-a}, \quad D = \frac{a-1}{q-a}.$$

Writing $m = v\ell + r$ with $0 \leq r < v$, the inequality defining $T_{a,q}(m)$ is

$$q^t \geq Ka^{v\ell+r} - D. \quad (9.22)$$

Set $t = u\ell + s$ and divide by M^ℓ to obtain

$$q^s \geq Ka^r - DM^{-\ell}. \quad (9.23)$$

For each residue r , let

$$s_r = \left\lceil \log_q(Ka^r) \right\rceil. \quad (9.24)$$

We have $q^{s_r-1} < Ka^r \leq q^{s_r}$. The positive distance between Ka^r and q^{s_r-1} is fixed, whereas $DM^{-\ell} \rightarrow 0$. Thus, for all sufficiently large ℓ , the least integer s satisfying (9.23) is exactly s_r . This remains true when Ka^r itself is an exact power of q ; the strict lower inequality is still available. Therefore

$$T_{a,q}(v\ell + r) = u\ell + s_r \quad \text{for all sufficiently large } \ell, \quad (9.25)$$

and, uniformly over the finitely many residues,

$$T_{a,q}(m + v) = T_{a,q}(m) + u \quad \text{for all sufficiently large } m. \quad (9.26)$$

Because $\log_q a < 1$, the maxima in (9.13) no longer truncate for either $m = n$ or $m = n + 1$ once n is large enough. Increasing the full product length by $2v$ increases both rectangular indices by v ; equation (9.26) then increases the exponent of a in (9.13) by $v - u$. This proves (9.5) on both parities. Multiplying $F_{a,q}(t)$ by $1 - a^{v-u}t^{2v}$ leaves only finitely many nonzero coefficients, so it is a polynomial and $F_{a,q}$ is rational; restricting (9.5) to even N gives $v_{a,q}(2n + 2v) = a^{v-u}v_{a,q}(2n)$ eventually, so $A_{a,q}$ is rational too. This proves (i) $\Rightarrow$ (ii) and (i) $\Rightarrow$ (iv). Together with (ii) $\Rightarrow$ (iii), (iv) $\Rightarrow$ (v), (v) $\Rightarrow$ (iii) and (iii) $\Rightarrow$ (i) from Subsection 9.2, every statement in Theorem 9.1 is now equivalent to every other, which completes its proof.

9.4 The even–odd identity for a general pair

Section 6 proved the even–odd identity for (2, 3). The general version is what makes the transfer between $A_{a,q}$ and $F_{a,q}$ above concrete, and it isolates precisely which initial indices are exceptional.

Proposition 9.3 (General even–odd identity). *Let $2 \leq a < q$ and set*

$$\mathcal{E}_{a,q} = \{n \geq 1 : T_{a,q}(n + 1) > n\} = \{n \geq 1 : g_n < a^{n+1}\}. \quad (9.27)$$

Then $\mathcal{E}_{a,q}$ is finite, and for $n \geq 1$

$$v_{a,q}(2n + 1) = \begin{cases} \frac{v_{a,q}(2n + 2)}{a}, & n \notin \mathcal{E}_{a,q}, \\ 2, & n \in \mathcal{E}_{a,q}. \end{cases} \quad (9.28)$$

At $n = 0$ one has $v_{a,q}(1) = a$ while $v_{a,q}(2) = 2$.

Proof. An odd prefix of length $2n + 1$ has $m = n + 1$ dense factors and n sparse ones, so $v_{a,q}(2n + 1) = U_{a,q}(n + 1, n)$ and $v_{a,q}(2n + 2) = U_{a,q}(n + 1, n + 1)$. Put $T = T_{a,q}(n + 1)$. If $T \leq n$, then (9.13) gives $2a^{n-T}$ and $2a^{n+1-T}$ respectively, whose ratio is a . If $T \geq n + 1$, both maxima vanish and both values equal 2. By the second equality in (9.11), $T_{a,q}(n + 1) > n$ is exactly $g_n < a^{n+1}$, giving the second description in (9.27). Finiteness follows from (9.20): since $\log_q a < 1$, the upper bound $\left\lceil (n + 1) \log_q a + \log_q(1/c) \right\rceil$ is at most n for all large n . Finally $U_{a,q}(1, 0) = a$ and $U_{a,q}(1, 1) = 2a^{\max(0, 1-T_{a,q}(1))} = 2$, because $T_{a,q}(1) \geq 1$. $\square$

For $a = 2$ the exceptional set can be listed completely.

Corollary 9.4 (Binary digit factors). *Let $a = 2$ and $q \geq 3$. Then*

$$\mathcal{E}_{2,q} = \begin{cases} \{1, 2, 3\}, & q = 3, \\ \{1\}, & q = 4, \\ \emptyset, & q \geq 5, \end{cases} \quad (9.29)$$

and consequently

$$F_{2,q}(t) = A_{2,q}(t^2) + \frac{A_{2,q}(t^2) - 1}{2t} + E_q(t), \quad E_q(t) = t + \mathbf{1}_{\{q \leq 4\}} t^3 + \mathbf{1}_{\{q=3\}} (t^5 + t^7). \quad (9.30)$$

The division by t is harmless because $A_{2,q}(t^2) - 1$ is divisible by t^2 as a formal power series.

Proof. By (9.27) we must decide when $g_n < 2^{n+1}$, where $g_n = ((q-2)q^n + 1)/(q-1)$ is increasing in q for fixed n . At $n = 1$, $g_1 = q - 1$, and $q - 1 < 4$ holds exactly for $q = 3, 4$. At $n = 2$, $g_2 = ((q-2)q^2 + 1)/(q-1)$ equals 5 for $q = 3$ and 11 for $q = 4$, so $g_2 < 8$ holds exactly for $q = 3$. At $n = 3$, g_3 equals 14 for $q = 3$ and 43 for $q = 4$, so $g_3 < 16$ holds exactly for $q = 3$. At $n = 4$ and $q = 3$, $g_4 = 41 \geq 32$; monotonicity in q extends this to all $q \geq 3$, and (3.9) propagates $g_n \geq 2^{n+1}$ to every larger n . This proves (9.29).

For the generating function, note that at every $n \in \mathcal{E}_{2,q}$ the inequality $T_{2,q}(n+1) \geq n+1$ forces $v_{2,q}(2n+2) = 2$, so (9.28) can be written uniformly for $n \geq 0$ as

$$v_{2,q}(2n+1) = \frac{v_{2,q}(2n+2)}{2} + \mathbf{1}_{\{n \in \{0\} \cup \mathcal{E}_{2,q}\}},$$

the case $n = 0$ being $2 = 1 + 1$. Multiplying by t^{2n+1} and summing, and using $v_{2,q}(0) = 1$ together with

$$\sum_{n \geq 0} v_{2,q}(2n+2) t^{2n+1} = \frac{A_{2,q}(t^2) - 1}{t},$$

gives the odd part $(A_{2,q}(t^2) - 1)/(2t) + E_q(t)$, where E_q collects the monomials t^{2n+1} for $n \in \{0\} \cup \mathcal{E}_{2,q}$ and therefore takes the stated form. Adding the even part $A_{2,q}(t^2)$ gives (9.30). $\square$

At $q = 3$ this is Proposition 6.2: the exceptional indices $\{0\} \cup \mathcal{E}_{2,3} = \{0, 1, 2, 3\}$ are the correction $\mathbf{1}_{\{0 \leq n \leq 3\}}$ of (6.4), and $E_3(t) = t + t^3 + t^5 + t^7$ matches (6.5). We do not assert an equally explicit list of exceptional indices for $a \geq 3$; Proposition 9.3 gives the structure and the finiteness, and the verification suite computes $\mathcal{E}_{a,q}$ directly for the ranges it tests.

9.5 Rational controls and explicit examples

The classification is not simply a separation between equal and unequal bases. Different bases can be dependent: for example, $4^3 = 8^2$. Table 2 records several eventual recurrences used as computational controls.

Table 2: Dependent-base controls. Each displayed recurrence is eventually valid; the verification suite checks every $40 \leq N \leq 200$.

a	q	Relation	Eventual recurrence
2	4	$4 = 2^2$	$v_{2,4}(N+4) = 2v_{2,4}(N)$
2	8	$8 = 2^3$	$v_{2,8}(N+6) = 4v_{2,8}(N)$
3	9	$9 = 3^2$	$v_{3,9}(N+4) = 3v_{3,9}(N)$
4	8	$8^2 = 4^3$	$v_{4,8}(N+6) = 4v_{4,8}(N)$
4	16	$16 = 4^2$	$v_{4,16}(N+4) = 4v_{4,16}(N)$
8	16	$16^3 = 8^4$	$v_{8,16}(N+8) = 8v_{8,16}(N)$
9	27	$27^2 = 9^3$	$v_{9,27}(N+6) = 9v_{9,27}(N)$

For a particularly simple closed form, take $a = 2$ and $q = 2^s$ with $s \geq 2$. For $n \geq 1$, writing $n = s\ell + r$, $0 \leq r < s$, one has $g_\ell < 2^n \leq g_{\ell+1}$. Indeed, $g_\ell < q^\ell \leq 2^n$ when $\ell \geq 1$, while $g_0 = 1 < 2^n$ when $\ell = 0$; and

$$g_{\ell+1} = \frac{(q-2)q^{\ell+1} + 1}{q-1} \geq \frac{q^{\ell+1}}{2} \geq 2^n.$$

It follows that $T_{2,2^s}(n) = \lfloor n/s \rfloor + 1$ and

$$v_{2,2^s}(2n) = 2^{n-\lfloor n/s \rfloor} \quad (n \geq 0). \quad (9.31)$$

Summing by residue classes gives

$$A_{2,2^s}(z) = \sum_{n \geq 0} v_{2,2^s}(2n) z^n = \frac{\sum_{j=0}^{s-1} 2^j z^j}{1 - 2^{s-1} z^s}. \quad (9.32)$$

Equivalently, (9.31) gives the clean shift

$$v_{2,2^s}(2(n+s)) = 2^{s-1} v_{2,2^s}(2n) \quad (n \geq 0), \quad (9.33)$$

which is (9.5) at $(a, q, u, v) = (2, 2^s, 1, s)$, valid here from $n = 0$ rather than merely eventually. The first three instances of (9.32) are

$$A_{2,4}(z) = \frac{1+2z}{1-2z^2}, \quad A_{2,8}(z) = \frac{1+2z+4z^2}{1-4z^3}, \quad A_{2,16}(z) = \frac{1+2z+4z^2+8z^3}{1-8z^4}, \quad (9.34)$$

the first two of which are the $(a, q) = (2, 4)$ and $(2, 8)$ rows of Table 2; combining them with Corollary 9.4 makes $F_{2,2^s}$ rational explicitly. In contrast, every pair $(2, q)$ with $q > 2$ not a power of 2 gives a counterexample with the original binary factors $1 + x^{G_i}$, and by Theorem 9.1 the resulting series is not even D-finite.

10 Reproducible exact computations

The proof does not infer an infinite statement from a long prefix. The computations serve three narrower purposes: validating the indexing and exceptional cases, checking the finite combinatorial identities by independent representations, and making the derived integer sequences easy to reuse.

10.1 Several distinct computational representations

The module `code/mixed_base.py` implements five routes. Direct multiplication stores every coefficient and multiplies by the digit factors from their definition. An endpoint sweep independently enumerates the sparse digit support and adds +1 at each interval start and -1 at each interval end; it computes the histogram of *all* coefficient values. The closed formula computes only the threshold and the resulting power of a . A streaming generator realizes the nonlinear recursion (10.1), carrying the triple (t_n, h_n, v_n) forward with one comparison per term and never recomputing a threshold from scratch. Finally, an odd-index route evaluates $v_{a,q}(2n+1)$ from the general even-odd identity (9.28) instead of from the odd rectangle. The first method does not use the interval model, the second does not use the carry-gap counting formula, the fourth does not use the closed threshold, and for $(2, 3)$ a sixth function evaluates the floor formula through the unrelated comparison $3^s \geq 2^{n+1}$ (Section 10.3).

The checker compares entire histograms when feasible, not merely their coefficient-one totals. It also checks total degree and the mass identity

$$\sum_s [x^s] Q_{m,n}^{a,q}(x) = Q_{m,n}^{a,q}(1) = a^{m+n}.$$

The zero-coefficient histogram entry refers only to exponents between zero and the polynomial's degree; infinitely many trailing zero coefficients are never counted.

Table 3: Executed finite verification. All decisions use exact integers.

Check	Instances
Dense rectangle: direct histogram = sweep histogram; count = formula	625
Larger binary-ternary rectangle: sweep count and mass identities	384
Independent interval-sweep grid, $a = 2$ and $3 \leq q \leq 12$	1 210
Seeded randomized interval-sweep cases	150
Enumerated carry-gap multiplicities, adjacency, and ruler identity	150
Ruler identity alone, $a = 2$ and $3 \leq q \leq 15$	156
Actual interleaved prefixes $P_0, \dots, P_{28}$	29
Threshold compared with both adjacent integer powers	149 050
Exact floor-formula identities, $0 \leq n \leq 10\,000$	10 001
Independent ternary closed form, $0 \leq n \leq 2\,000$	2 001
Input recurrence (1.5)	1 000
Even-odd identity (6.4)	1 000
Binary-family thresholds and quotient set, $3 \leq q \leq 64$	18 600
Streaming recurrence (10.1) against the closed formula	37 262
General even-odd identity (9.28), $a = 2$	37 262
General even-odd identity and streaming, eight pairs with $a \geq 3$	1 608
Exponent recurrence (9.10), $3 \leq q \leq 64$	4 774
Power-of-two rational subfamily (9.31)	5 409
Dependent-family recurrence, seven families and 161 indices each	1 127

The dense grid uses $2 \leq a \leq 6$, $a + 1 \leq q \leq a + 5$, and $0 \leq m, n \leq 4$. The larger sweep grid uses $(a, q) = (2, 3)$, $1 \leq m \leq 24$, and $1 \leq n \leq 16$. The independent sweep grid uses $a = 2$, $3 \leq q \leq 12$ and $0 \leq m, n \leq 10$, and includes the degenerate cases $m = 0$ and $n = 0$ even though (9.13) assumes both positive. The randomized cases are drawn from the fixed seed 20260919 with $3 \leq q \leq 40$, $0 \leq m \leq 30$ and $0 \leq n \leq 15$; the seed is recorded in the result file so that the identical cases are reproduced on every run. The carry checks enumerate $1 \leq n \leq 6$ for the same 25 base pairs as the dense grid, and additionally verify (9.9) at every single index rather than only in aggregate; the wider ruler check uses $a = 2$, $3 \leq q \leq 15$ and $1 \leq n \leq 12$. The threshold tests cover $0 \leq m \leq 100\,000$ for $(2, 3)$ and $0 \leq m \leq 1000$ for 49 additional enumerated base-pair cases ($2 \leq a \leq 8$ and $a + 1 \leq q \leq a + 7$; this includes $(2, 3)$ again). The binary-family checks use $3 \leq q \leq 64$ with $1 \leq n \leq 300$ for the thresholds and $0 \leq N \leq 600$ for the parity identity and the streaming recurrence; the eight pairs with $a \geq 3$ are $(3, 4)$, $(3, 5)$, $(3, 7)$, $(3, 9)$, $(4, 5)$, $(4, 9)$, $(5, 7)$, $(6, 7)$ over $0 \leq N \leq 200$. For each of those pairs the

exceptional set $\mathcal{E}_{a,q}$ of (9.27) is computed directly and found to be a finite initial run, as Proposition 9.3 requires; for $a = 2$ the computed sets match (9.29) exactly. The largest directly expanded polynomial has degree 2 407 867, reached by the full prefix P_{28} .

The supplied run completed successfully in about 19 seconds. The file `data/verification.json` records the counts, the actual elapsed time, the random seed, the computed exceptional sets, and a SHA-256 digest of the dense-grid histograms. Timing is an observation about the supplied execution environment, not a performance guarantee.

10.2 A compact formula and an exact streaming recurrence

The following compact function computes any full-prefix count in the family. It is an executable form of Proposition 9.2.

```
def count_one_prefix(N: int, a: int = 2, q: int = 3) -> int:
    if N < 0 or not 2 <= a < q:
        raise ValueError("Need N >= 0 and integer bases 2 <= a < q")
    m, n = (N + 1) // 2, N // 2
    if n == 0:
        return a**m
    if m == 0:
        return a**n
    t, q_power = 0, 1
    target = (q - 1) * a**m
    while (q - a) * q_power + a - 1 < target:
        q_power *= q
        t += 1
    return 2 * a**max(0, n - t)
```

The full module also offers a streaming threshold computation that reuses a^m and $q^{T(m)}$. The counts themselves can be streamed by the same device, with one exact comparison per new term.

Proposition 10.1 (Exact streaming recurrence). *Fix $2 \leq a < q$ and write $t_n = T_{a,q}(n)$, $h_n = g_{t_n}$ and $v_n = v_{a,q}(2n)$. Suppose $n \geq 1$ and $t_n \leq n$. Then*

$$(t_{n+1}, h_{n+1}, v_{n+1}) = \begin{cases} (t_n + 1, qh_n - (a - 1), v_n), & h_n < a^{n+1}, \\ (t_n, h_n, a v_n), & h_n \geq a^{n+1}. \end{cases} \quad (10.1)$$

For $a = 2$ and any $q \geq 3$ the side condition $t_n \leq n$ holds for every $n \geq 1$ by (3.9), and the recursion may be seeded at $t_1 = 1$, $h_1 = q - 1$, $v_1 = 2$.

Proof. By (9.16), $t_{n+1} - t_n \in \{0, 1\}$. It is 0 exactly when $h_n = g_{t_n} \geq a^{n+1}$, and otherwise 1, in which case $h_{n+1} = g_{t_{n+1}} = qh_n - (a - 1)$ by (9.15). Since $t_n \leq n$, both maxima in (9.21) are attained by their second argument, so $v_n = 2a^{n-t_n}$ and $v_{n+1} = 2a^{n+1-t_{n+1}}$; the two cases give $v_{n+1} = v_n$ and $v_{n+1} = av_n$ respectively. For $a = 2$, (3.9) gives $g_n \geq 2^n$, so $t_n \leq n$; and $g_1 = ((q - 2)q + 1)/(q - 1) = q - 1 \geq 2$ with $g_0 = 1 < 2$ gives $t_1 = 1$ and $h_1 = q - 1$. $\square$

Equation (10.1) is a nonlinear recursion — the branch depends on an inequality between the state and a^{n+1} — and it is completely elementary. That is worth stating plainly, because it is easy to read “not P-recursive” as “hard to compute.” It is not. For fixed bases, streaming $v_0, \dots, v_N$ costs $O(N)$ integer updates and $O(N)$ bits of working storage: every stored integer has $O(N)$ bits, and multiplication by

the fixed base, subtraction, comparison, and multiplication of the running count by a each have linear bit cost. Summed over the prefix this is

$$O(N^2) \text{ bit operations,} \quad O(N) \text{ working bits.} \quad (10.2)$$

The exponent 2 cannot be improved if the results are actually written out: by (9.20) the exponent of a in v_n grows with positive limiting slope $1 - \log_q a > 0$, so the binary representations of $v_0, \dots, v_N$ already occupy $\Theta(N^2)$ bits. The simplicity of this threshold recursion is fully compatible with the impossibility of a linear recurrence with polynomial coefficients; the two statements are about different classes of algorithm.

For a single rectangle, direct dense expansion has a degree that grows exponentially in the sparse-digit length. The sweep instead uses $O(a^n)$ endpoint events and the implementation sorts them in $O(a^n \log(a^n))$ comparisons. The formula avoids both enumerations. These counts describe arithmetic or comparison operations; they do not treat arbitrarily large integer operations as constant-time in the bit-complexity discussion.

10.3 Why floating-point floors are avoided

The closed forms (1.7) and (9.12) are written with floors and ceilings of logarithms, and it would be natural to implement them that way. The module never does, and the reason is not stylistic.

The hazard is that $(n+1)\alpha$ can come very close to an integer. The decision $\lfloor (n+1)\alpha \rfloor$ is discontinuous exactly there, so a double-precision evaluation of α and a multiplication — each correct to a relative error of order 2^{-53} — can land on the wrong side of the discontinuity once n is large, and does so without any warning. How large is a question about the irrationality measure of $\log_3 2$, which is precisely the kind of question this report avoids depending on. The exported sequence values are therefore decided by exact integer comparisons only, and no numerical approximation to a logarithm enters any correctness assertion anywhere in the package.

Concretely, two genuinely independent integer formulations are implemented and cross-checked against each other:

- the gap threshold, $T_{a,q}(m) = \min\{t \geq 0 : g_t \geq a^m\}$, evaluated by the recursion $g_{t+1} = qg_t - (a-1)$ of (9.15);
- for $(2, 3)$ only, the least s with $3^s \geq 2^{n+1}$, evaluated by repeated multiplication by three and a single comparison.

The second route never forms a gap and never uses (9.15); the first never forms a power of three in isolation. They agree on $0 \leq n \leq 2000$, which is the row “Independent ternary closed form” of Table 3. The near-integer hazard is not merely sidestepped by the choice of representation: there is nothing to round.

10.4 Package contents and reproduction

Run the verification suite from the project directory with

```
python code/verify.py
```

It requires only the Python standard library and is written for Python 3.9 or later. Do not run it with `-O`, which disables assertions. The supplied verification was executed using Python 3.14. A reduced

run is available as `python code/verify.py --quick`; the full run is the one reported in Table 3 and is the one to prefer. A Makefile provides `make test`, `make pdf`, `make all` and `make clean` targets. The command

```
python code/mixed_base.py 40 --a 2 --q 3
```

prints the first 41 full-prefix counts, and the module can be imported directly:

```
from mixed_base import interleaved_count, count_one, even_count_stream

assert interleaved_count(2, 3, 20) == 16
assert count_one(2, 3, 10, 10) == 16
assert list(even_count_stream(2, 3, 10)) == [1, 2, 2, 2, 2, 4, 4, 4, 8, 8, 16]
```

The CSV files contain indices 0 through 1000; a separate plain-text file uses the familiar two-column sequence format, and `data/base_comparison.csv` places the dependent and independent cases side by side for eight base pairs. No OEIS identifier is assigned or claimed.

The PDF is built with two passes of `pdflatex article.tex`; all needed figure PDFs are included. Recreating the figures with `python code/make_figures.py` additionally requires Matplotlib. The plots use floating-point arithmetic only for visualization; all verification decisions and all exported integer sequence values are exact. The README gives the complete directory map and reproduction commands.

11 What is settled, and what remains

The universal implication in (1.1)–(1.2) is false. The counterexample uses positive integer exponents tending to infinity, a fourth-order constant-coefficient recurrence for those exponents, fixed binary factors, and the simplest nonzero coefficient value $k = 1$. The output generating function not only fails to be rational: it has a natural boundary and is not D-finite. Within the specified two-base digit family, Theorem 9.1 gives a complete classification of rationality and of D-finiteness together.

No search through enormous exceptional indices is needed to witness the failure. The obstruction is visible at every scale at once: it is the irrational frequency of exact doublings, recorded in the Sturmian word (6.3). Interleaving two geometric sequences preserves C-finiteness of the exponents — (9.10) is an order-four constant-coefficient recurrence for every pair — but the operation “count the coefficients equal to one” need not preserve even P-recursiveness of the result. Nor, by Proposition 10.1, does the failure of P-recursiveness make the sequence hard to compute.

Four boundaries of this result should be kept separate. First, the exponents in the counterexample are not eventually increasing: concretely, $G_6 = 9$ is followed by $G_7 = 8$. The same is true of the general interleaving when $a < q$: eventually $q^j > a^{j+1}$, so each sufficiently late even-to-odd step decreases. This is not a violation of the stated hypotheses, which require only positivity and divergence. Reordering all exponents would preserve a finite product with a fixed multiset of factors, but it changes the prefix sequence and need not preserve the rationality of the exponent generating function. Thus this construction does not settle a separately imposed monotone version. For the same reason, no conclusion is asserted here for a version of the question restricted to a single simple dominant characteristic root, or to exponent recurrences with prescribed positive coefficients.

Second, the proofs count coefficients equal to one. They do not establish a corresponding classification for every fixed $k > 1$. For those values, multiple neighboring intervals and their overlaps

matter; the isolated-gap identity alone no longer determines the answer. This is not out of reach: the event-sweep verifier already computes the complete multiplicity histograms, so exact formulas for fixed $k > 1$ have a concrete starting point. It is simply not claimed here that every fixed- k series in this family has the same classification.

Third, the classification concerns the family $G_{2j+1} = a^j$, $G_{2j+2} = q^j$ with digit set $0, \dots, a-1$. It is not a classification of all C-finite exponent sequences. The arithmetic mechanism suggests studying which recurrence families force periodic relative scales and which allow irrational threshold frequencies, but no claim about all such families is made here. What the family does supply is a necessary warning for any proposed general theorem: rationality of the exponent series is by itself too weak a hypothesis. A useful sufficient condition would have to control the relative digit scales, or the possible carry states, and not merely the existence of a linear recurrence for the exponents.

Fourth, on novelty. Interval counting, finite-state recurrence arguments, reduction modulo a prime, and irrational rotation averages are standard, and none of them is claimed as new. The intended contribution is their explicit combination into the counterexample, the exact enumeration, the two independent algebraic obstructions, the natural boundary, and the two-base classification proved here.

These are directions left untouched by this report, not assertions that each associated broader question has independently been verified to be open in the literature. The completed result is the explicit counterexample, its natural-boundary strengthening, and the exact two-base classification.

References

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- [3] Richard P. Stanley, *Some linear recurrences motivated by Stern’s diatomic array*, American Mathematical Monthly **127** (2020), 99–111. Preprint: <https://arxiv.org/abs/1901.04647>.
- [4] Richard P. Stanley, *Theorems and conjectures on some rational generating functions*, European Journal of Combinatorics **119** (2024), article 103814. <https://doi.org/10.1016/j.ejc.2023.103814>. Preprint: arXiv:2101.02131, version 3, September 30, 2021, 25 pp., <https://arxiv.org/abs/2101.02131>.
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